

Why Event Density's Breadth Is Evidence, Not Bookkeeping

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The claim

Event Density reproduces the *form and mechanism* of physics across twelve separate, currently-unrelated domains, some 159 tiered results, from thirteen substrate primitives, ten constants, and fifteen postulates. Thirty-eight inputs, none of which are dialable.

The room in a mature science to predict what the incumbents cannot test is nearly full. While ED does make novel predictions, what it claims is narrower and, taken seriously, harder to dismiss: **one structure gives the shape of the whole landscape, where the standard picture needs several mutually-incompatible theories and their own set of brute inputs.**

The objection, met head-on

The reflex is immediate and fair to state: *“So what? You fit facts that were already known. Any flexible theory can be reverse-engineered to match what we already measured.”*

That reflex is a **proxy**, and it's aimed at the wrong target. It treats “known versus new” as the thing that matters. It isn't. A fact confirms a theory to the degree it would be *surprising unless the theory were true*. Whether the fact was discovered before or after the theory never enters that reasoning. The importance of *when* a fact is discovered is psychological, not logical. What the “you just fit it” reflex is *actually* reaching for is a real and legitimate question:

Could you have tuned your theory to fit this?

A new prediction avoids this question: the fact didn't exist before the theory was built, so the theory couldn't have been fit. But novelty is *one* way to guarantee no-fitting, not the only one. A known fact reproduced by a structure too rigid to have been fitted to it carries just as much weight. So the real question is not old-versus-new. It is **tunable-versus-untunable**. And that is a question you can actually answer.

The untunability argument

Two moves, and the whole case for taking Event Density seriously rests on them.

1. The ratio. Few knobs, many outputs. “This explains everything” from a theory with a thousand free parameters is worth nothing, because a thousand-knob theory fits anything. The same words from a theory with *thirty-eight* fixed inputs are worth a great deal, *because you*

cannot tune thirty-eight things to independently hit gravity and the Born rule and viscoelastic relaxation and the dark sector by accident. The figure of merit for a unified theory is:

independent results reproduced, divided by free inputs.

For Event Density that ratio is 159 results over 38 inputs, better than four to one. The same accounting goes badly for the standard picture, because “GR + QM + a dark-matter particle + brute constants” is four theories in a trenchcoat, each with its own inputs, two of them not even compatible (they coarse-grain time in different ways).

An honest footnote, because a skeptic will reach for it: of those thirty-eight, the ten constants are *inherited values* (c , $\hbar$, ℓ_P , H_0 , the masses). Those are real, and they are not zero. But they are not a fresh knob per domain; they are the *handful* of scales that recur everywhere. ℓ_P sets Newton’s constant, the area law, and the Planck mass; H_0 sets the MOND scale and the cosmological constant. So the value layer is shared and constrained too, not tunable domain by domain. The rigidity holds at every level. (The Standard Model of particle physics inherits its constants the same way, and nobody counts that against it.)

2. The shared structure. Fit one, and the rest are forced. The reproductions do not each get their own private derivation. They run through the *same* thirteen primitives, so fixing the structure in one domain constrains what it does in all the others. The cleanest specimen is forced by the arrow of commitment itself, the single causal cone it makes matter and gravity share:

From that one cone, two things are forced at once. Gravitational waves travel at exactly the speed of light, $c_{GW} = c$, the identity GW170817 *confirmed* to one part in 10^{15} , and which killed a dozen rival gravity theories in a single measurement. And the preferred-frame parameter $\alpha_2 = 0$, *exactly*, a sharp null that lunar-laser-ranging and pulsar timing can check. These are not two guesses; they are one cone seen twice. You cannot have the confirmed one without the forced one, because the same structure sets both. The day α_2 is measured nonzero, ED is dead, $c_{GW} = c$ passing notwithstanding.

Honest tier: $c_{GW} = c$ is *shared* with General Relativity (GR forces it too), so it is not evidence for ED alone. The point is not that ED owns it, but that ED cannot *separate* it from $\alpha_2 = 0$. That inseparability is the untunability: one structure, forced by the arrow, that hands you a confirmed identity and a distinctive sharp null on the same string, with no dial between them.

Untunability also lands in single numbers. The oldest: fit gravity to the everyday limit and you have set how it warps *time*; the same structure then *forces* how it warps *space*, so starlight must bend by exactly twice the naive value, the 1.75 arcseconds Eddington measured in 1919. You cannot fit the apple and then dial the starlight. And from cosmology: the MOND scale, $a_0 = cH_0/2\pi$, no free coefficient, just the speed of light times the Hubble rate over 2π , landing within about ten percent of the measured value. No knob to turn: you get that number from the structure or you don’t.

The sample ledger (representative, tiered, checkable)

Not exhaustive, and deliberately spanning *unrelated* fields, because the breadth across independent domains is the point. Each row’s honest tier lives in its own paper; the tiers here are

conservative.

Domain (independent)	What ED reproduces	Tier of the <i>form</i>	The <i>value</i>
Weak-field gravity	Schwarzschild metric, Mercury precession, factor-2 light bending, redshift	Derived (conditional on one labeled postulate)	$c, \hbar, \ell_P$ inherited
Newtonian gravity	Inverse-square law	Derived , two independent routes	—
Newton's constant	$G = c^3 \ell_P^2 / \hbar$, the dimension living in the area factor	Derived form	no new input — set by $c, \ell_P, \hbar$
Covariant gravity	Einstein- <i>form</i> field equation (khronometric, not pure GR)	Grounded (thermodynamic route)	κ inherited; Λ via H_0 ($w=-1$ forced, smallness open)
Galactic rotation	MOND law (interference cross-term = the mechanism); $a_0 =$ $cH_0/2\pi$ ($\sim 10\%$); baryonic Tully-Fisher	a_0 form derived; BTFR conditional on one postulate	a_0 value inherited via H_0
Quantum probability	Born rule, non-contextuality, from intrinsic bandwidth + commitment	Reconstructed (account-tier)	residual: Solèr rigor
Quantum kinematics	Complex Hilbert space ($\mathbb{C}$ <i>selected</i> from the Solèr menu), orthogonality from distinguishability	Selected / reduced	—
Soft matter (rheology)	Maxwell viscoelastic stress-relaxation	Derived form (V5 + coarse-graining)	relaxation time inherited
Black-hole entropy	Area law $S \propto A$ (severed-edge count, on an assumed geometry)	Scaling measured, half-structural	the 1/4 inherited (thermal)
Three dimensions	3 selected by 1D-linking and the holographic count	Structural bridge	postulated at bottom; exactly-3 open

Read the tier column, not just the domain column. Leading with it, rather than burying it, is the whole discipline of the case.

What this is NOT

- It is **not** “ED derives the numbers.” Most values are inherited. What is unified is the *form and mechanism*, not the constants. Say that first, always.
- It is **not** out-predicting GR or QM in the tested regime. There is almost no room there, and that is honest.
- It is **not** unfalsifiable. There is a companion document, *22 Ways to Kill Event Density (+ Eight More)*, that is nothing but **thirty numbered kill-conditions**, each a specific number, sign, or shape with a stated way to fail it: $\alpha_2 = 0$ *exactly, zero* intrinsic scatter in the baryonic Tully-Fisher relation, dark energy $w = -1$ with no drift, a sharp architecture-independent decoherence wall, and twenty-six more. Every one is a number someone can go measure.
- It is **not** proof. Consilience *persuades*; it does not *compel*. A determined skeptic can still say “you accommodate the known facts, so what,” and the only thing that closes that door is a confirmed novel prediction. ED has two live, non-Planck-suppressed candidates that are not postdictions: the redshift evolution of the MOND scale, $a_0(z) = cH(z)/2\pi$ (its direction already confirmed at about 30σ , killing constant- a_0 MOND, though its exact rate is under mild tension in the first data), and the sharp quantum-to-classical boundary (honestly still a shape-claim, not a number). Consilience is the body of the case; those predictions are the intended knockout. Both, not either.

Bottom line

The breadth of explanation from thirteen primitives is the argument, and it holds. Not because old facts secretly count as new, they always counted. Because **one rigid structure reproducing many independent domains cannot be a coincidence and cannot be reverse-engineered**. A theory you *could* have fitted to the data explains nothing by matching it. A theory too small to have been fitted, that matches anyway, across gravity, quantum mechanics, kinematics, and the dark sector, is telling you something about the world. That is the whole of the case, stated without inflation: reproduced from thirteen, tiered in the window, the fit itself the evidence.